

LAST NAME: _____

Practice Quiz 05C — MATH 2055 — Fall 2026

Practice for Quiz 05

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

A student claims that, for all real numbers x, y ,

$$x + y \neq 0 \longrightarrow (x \neq 0 \wedge y \neq 0).$$

The proposed proof says: *Suppose $x + y \neq 0$. For a contradiction, assume $x = 0$ and $y = 0$. Then $x + y = 0$, a contradiction. Therefore $x \neq 0$ and $y \neq 0$.*

1. Identify the incorrect negation and write the correct negation of the proposed conclusion.
2. Give real values of x, y that disprove the student's claim.